

Postmenopausal Bone Loss Is Prevented by Treatment with Low-Dosage Estrogen with Calcium

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Bone mass was measured prospectively in 73 women during the period immediately after menopause. By comparing the rates of loss at three skeletal sites, we assessed the protective effects of calcium supplements given alone or combined with low-dosage estrogen therapy. After 2 years of follow-up, spinal trabecular mineral content, measured by quantitative computed tomography, decreased by a mean of 9.0% ($p = 0.002$ compared with baseline) in untreated women and a mean of 10.5% ($p = 0.0001$) in women given calcium supplements alone. By contrast, women given conjugated estrogens, 0.3 mg/d, with calcium supplements showed an insignificant increase of 2.3%. Significant losses of a lesser magnitude were seen in the appendicular cortical skeleton of women not receiving therapy and in those receiving calcium alone, but no significant changes were observed in women receiving estrogen with calcium.

BONE MINERAL LOSS is known to accelerate after ovarian function ceases. Various quantitative bone mineral techniques have been used to assess postmenopausal changes at a number of skeletal sites. For the past decade, our group has developed and applied the technique of computed tomography to the quantitation of spinal trabecular mineral content. After completing our initial prospective study (1) of spinal mineral loss in oophorectomized women given placebo or graded doses of estrogen, we designed this study to examine the rate of bone loss in untreated women undergoing spontaneous menopause and to test for possible protective effects of estrogen or calcium.

Estrogen replacement therapy, begun near the time of menopause, prevents bone loss and attendant osteoporotic fractures. A high-calcium diet has been widely recommended to protect against postmenopausal bone loss and

fracture, but its efficacy is not well substantiated. We (1) and others (2-4) have studied the osteotrophic dose-response characteristics of various forms of estrogen in order to find the minimum effective dosage. Considerable evidence suggests that the minimum dosage that will prevent bone loss is 0.6 mg/d of conjugated estrogens, or the equivalent dosage of other estrogens. Using lower dosages could result in less endometrial stimulation and reduced gynecologic complications. In this study, we tested the effectiveness of 0.3 mg/d of conjugated estrogens in women having spontaneous menopause who were concurrently given calcium supplements; we hypothesized that the effects of these two therapies might be additive when combined and thus might better prevent bone loss. We also observed the rate of mineral loss in postmenopausal women taking calcium supplements without estrogen, as well as the rate in those receiving no therapy.

Methods

White, Asian, and Hispanic women between the ages of 43 and 59 years who were members of the Kaiser Permanente Health Plan in San Francisco were asked to enroll in a prospective study of menopausal bone loss. We solicited subjects by mail and subsequently screened all respondents. After being informed of the risks and benefits of each therapy offered, all entrants were allowed to select their treatment themselves. The study protocol was approved by the regional Institutional Review Board.

All women had had spontaneous menopause 6 to 36 months before entry into the study. Menopausal status was confirmed by testing for elevated levels of follicle-stimulating hormone (> 50 mIU/mL). We excluded any woman with clinical or laboratory evidence of bone disease and those with any medical condition that could predispose to bone loss. Low spinal mineral content (< 90 mg/cm³) was found at initial screening in eight women, and they were advised to take 0.625 mg/d of conjugated estrogens (Premarin; Ayerst Laboratories, New York, New York), a dosage known to be effective; they had the

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same follow-up studies, but their results were analyzed separately. None of the patients had previously used estrogen therapy.

TREATMENT

For both practical and ethical reasons, we allowed each woman to select her own method of osteoporosis prevention. The women who showed interest in our study were educated and well informed about osteoporosis and estrogen therapy. We surmised that most of them would be reluctant to accept random assignment, and we thought it unethical to assign women to no therapy and thereby deprive them of dietary calcium augmentation, the standard of practice in the community. Given proper informed consent, 25 women elected no special treatment and simply maintained their usual dietary calcium intake; 44 chose to supplement their dietary calcium intake up to 1500 mg/d by taking 1000 mg/d of oyster shell calcium (Os-Cal; Marion Laboratories, Kansas City, Missouri) in two divided doses; and 15 chose 0.3 mg/d of conjugated estrogens (Premarin) combined with the same calcium supplementation. All medications were supplied by the investigator and were dispensed at regular 3-month intervals.

Of the 84 women who entered the study, 11 dropped out before the first year and were not included in any analyses. Six women dropped out between the first-year and second-year studies, 3 from the calcium-only group and 3 from the estrogen group. Their decision to leave was due to lack of interest and not to side effects or bone mineral changes seen in the first year. Estrogen was given for 25 days of each month, and for most women a progestogen was added to the regimen from day 16 through day 25 (either medroxyprogesterone, 10 mg, or norethindrone acetate, 5 mg). At 3-month intervals, tablet counts were done to ensure adequate compliance. All recruiting was completed within a year, and before we knew any treatment results. The same general counseling was given to all women, and special attempts were made to avoid directing subjects into any treatment group. The women were aware of their baseline bone mineral values and could use that information in choosing their mode of treatment.

CLINICAL VARIABLES AND BONE MINERAL DETERMINATIONS

In addition to general demographic variables, we recorded each woman's parity, time since menopause, smoking history, and activity level. Activity was graded from 1 to 3: 1 = sedentary, 2 = active, and 3 = participation in some sports, active hobbies, or light exercise. No woman was practicing intense physical conditioning. Current consumption of at least half a pack of cigarettes daily was defined as smoking. Dietary calcium intake was assessed from responses to a food frequency questionnaire.

Bone mineral content at the radial diaphyseal site was measured by single-photon absorptiometry, according to the method of Cameron and coworkers (5). Industrial radiographs of the hands were examined with a 10-power loupe provided with a reticule, allowing quantitation of the mean combined cortical

Table 1. Comparisons of Baseline Variables in Postmenopausal Women by Treatment Group*

Variable	No Treatment	Calcium Alone	Estrogen with Calcium
Age, yrs	51.6 ± 2.8	53.0 ± 3.0	51.9 ± 1.9
Height, cm	162.3 ± 7.6	163.6 ± 8.1	163.3 ± 6.9
Weight, kg	65.6 ± 15.4	62.6 ± 9.1	59.0 ± 7.4
Months since menopause, n	22.6 ± 9.7	22.3 ± 10.2	21.5 ± 9.9
Activity grade†	1.8 ± 0.6	2.1 ± 0.6	2.1 ± 0.7
Smokers, %	26.3	5.1	6.7
Nulliparous, %	26.3	33.3	33.3
Descent, %			
White	68.4	82.0	80.0
Asian	21.0	15.4	20.0
Hispanic	10.5	2.6	0.0

* Data are mean ± SD where appropriate. None of the differences among the groups is significant.

† See Methods for description of grading scheme.

thickness of the second metacarpal midshafts. Quantitative computed tomography of the L1 and L2 vertebrae was used to measure spinal trabecular mineral content. These measurements were obtained at yearly intervals without knowledge of the patient's treatment status.

STATISTICAL ANALYSES

Intergroup differences of demographic variables were evaluated for significance by analysis of variance. Differences between each woman's baseline and subsequent bone mineral measurements were assessed by the paired *t*-test, and group comparisons were done by analysis of variance. The change in spinal trabecular mineral content after 2 years served as the continuous dependent variable in a stepwise multiple regression analysis of all clinical variables. These analyses were done at the computer center of the University of California, San Francisco, and used the Statistical Analysis System (SAS Institute, Inc., Cary, North Carolina).

Results

The treatment groups were similar in most baseline variables (Table 1), except that women receiving no treatment were 3 kg heavier than those receiving calcium supplements and were 6.6 kg heavier than those receiving estrogen (neither difference was significant). Baseline bone mineral values were close to those expected and were similar in all treatment groups (Table 2). The range of total calcium intake from diet and from any supplemental calcium was 258 to 994 mg in untreated women, 1144 to 2950 mg in those taking calcium alone, and 1463

Table 2. Comparisons of Baseline Bone Mineral Values by Treatment Group, Expressed as Absolute Value and Its Percentage of Expected for Age*

Assessment	No Treatment	Calcium Alone	Estrogen with Calcium
Metacarpal combined cortical thickness			
Millimeters	4.87 ± 0.62	4.66 ± 0.49	4.68 ± 0.62
Percentage	96.3 ± 13.6	93.0 ± 10.2	92.3 ± 12.7
Radial bone mineral content			
Grams/centimeter	0.701 ± 0.074	0.699 ± 0.062	0.688 ± 0.071
Percentage	97.8 ± 11.2	98.1 ± 9.2	93.8 ± 10.6
Spinal trabecular mineral content			
Milligrams/cubic centimeter	134.8 ± 29.3	119.9 ± 21.0	122.3 ± 32.8
Percentage	114.5 ± 18.0	101.8 ± 18.1	104.3 ± 29.4

* Data are mean ± SD. None of the differences among groups is significant.

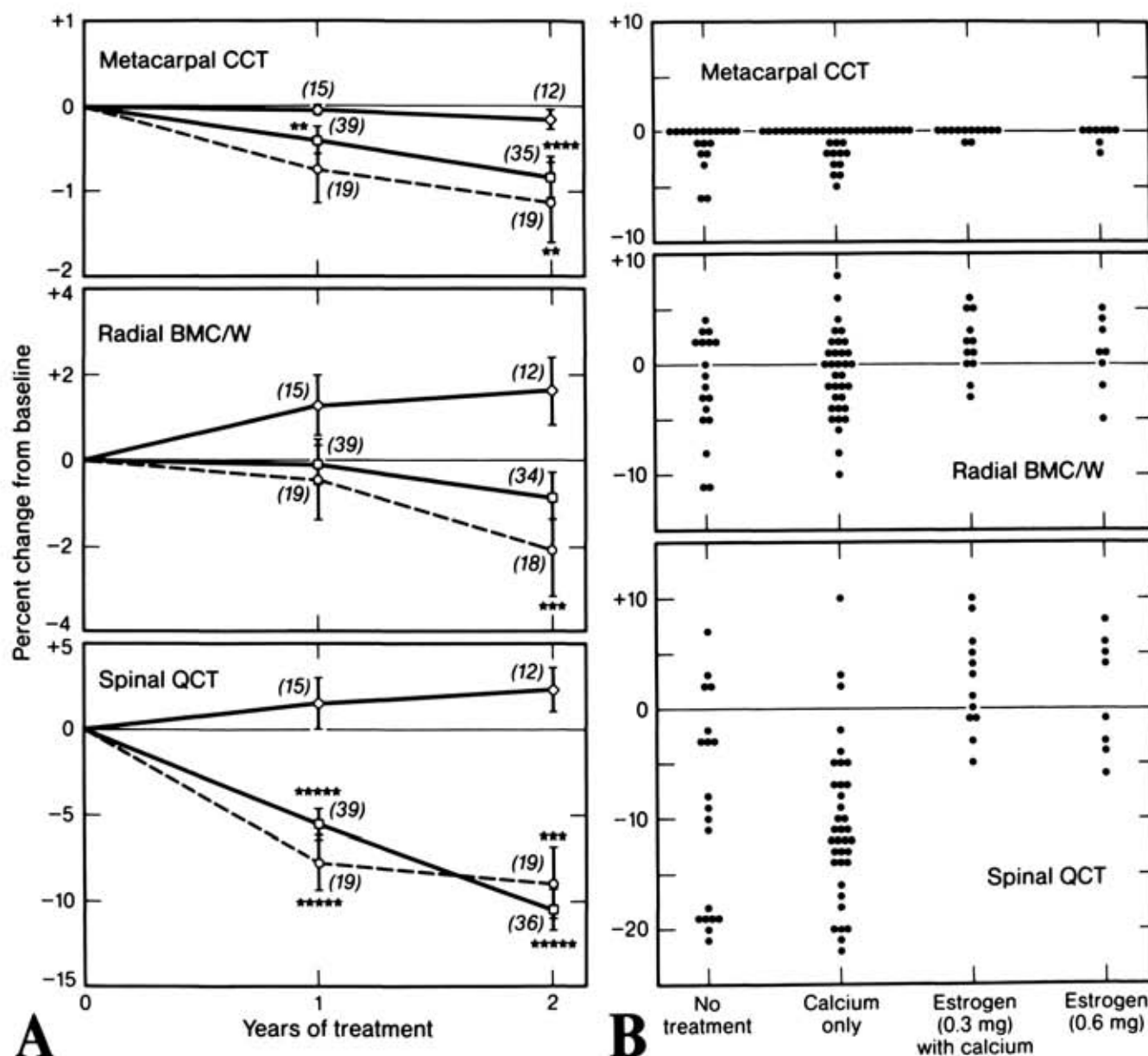


Figure 1A. Mean percent change from baseline (± 1 SE) in bone mineral assessments after 1 and 2 years: metacarpal combined cortical thickness (CCT), radial bone mineral content per width (BMC/W), and spinal trabecular mineral content measured by quantitative computed tomography (QCT). The open circles represent the no-treatment group; open squares, the calcium-only group; and open diamonds, the estrogen-and-calcium group. The asterisks indicate significant differences from the baseline values: two asterisks indicate $p = 0.02$; one asterisk, $p < 0.05$; four asterisks, $p = 0.001$; five asterisks, $p = 0.0001$; and three asterisks, $p = 0.002$. Numbers in parentheses are the numbers of women tested. **Figure 1B.** Each woman's skeletal changes after 2 years, expressed as a percentage of baseline values.

to 2498 mg in those taking calcium with estrogen. The mean daily calcium intakes ($\pm$ SD) were 662 ± 208 , 1863 ± 369 , and 1733 ± 279 mg in the three groups.

Figure 1A shows the mean bone mineral changes observed for the three bone tests after 1 and 2 years in each treatment group. Each woman's cumulative 2-year changes are shown in Figure 1B. After 2 years of follow-up, comparisons between the mean 2-year values and the mean baseline values showed a significant reduction of 9.0% in spinal trabecular mineral content ($p = 0.002$) and 1.1% in combined cortical thickness ($p = 0.02$) in women receiving no treatment and a decrease of 10.5% in spinal trabecular mineral content ($p = 0.0001$) and

0.8% in cortical thickness ($p = 0.001$) in women taking calcium supplements. The radial bone mineral content also was reduced by 2.1% ($p < 0.05$) for untreated women and 0.9% ($p > 0.2$) for calcium-treated women.

Women taking both estrogen and calcium showed no significant changes in spinal trabecular mineral content (+2.3%), radial bone mineral content (+1.6%), and metacarpal combined cortical thickness (-0.2%). The lower 95% confidence limits for each of these results provide a measure of the assurance that bone loss was not occurring. These limits were -0.3% for the spine, +0.1% for the radius, and -0.4% for the metacarpal. When compared with untreated women or with those

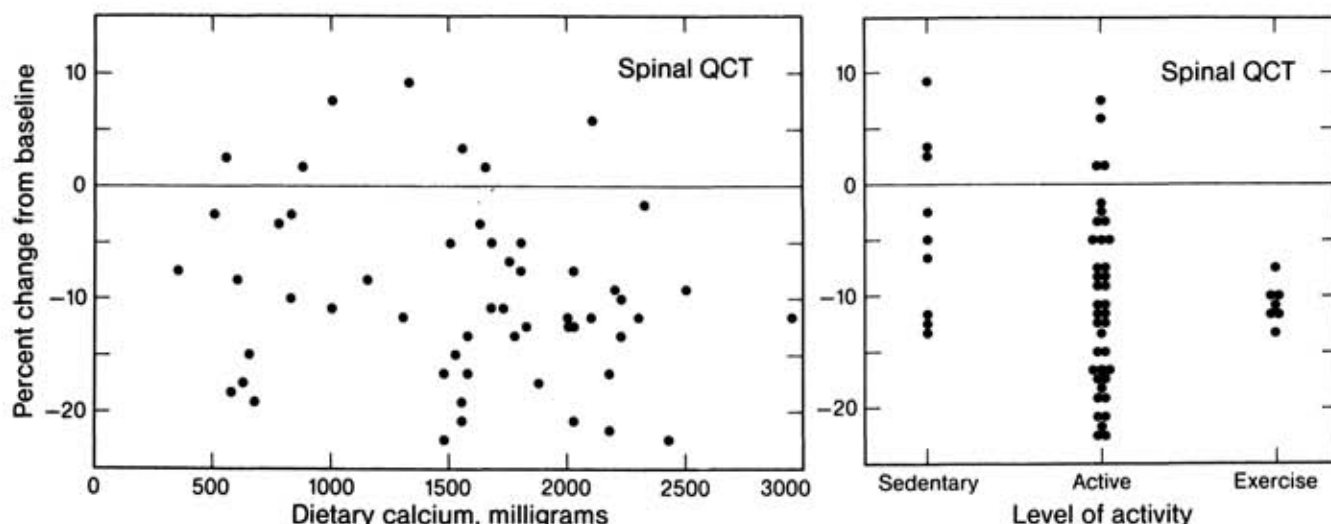


Figure 2. Each woman's percent change in spinal trabecular mineral content (as measured by quantitative computed tomography [QCT]) after 2 years compared with daily calcium intake (*left panel*) and activity level (*right panel*) (estrogen users excluded from both comparisons).

taking calcium alone at 2 years, women treated with both estrogen and calcium had significantly greater preservation of spinal trabecular mineral content ($p = 0.001$ for both) and radial mineral content ($p = 0.01$, $p < 0.02$), but differences in metacarpal thickness just failed to reach statistical significance ($p = 0.12$, $p = 0.09$).

When the data from untreated women were combined with data from those taking calcium and subsequently compared with data from those taking estrogen, significant differences were seen in all three bone mineral tests ($p < 0.0001$ for spinal trabecular mineral content, $p < 0.01$ for radial mineral content, and $p = 0.05$ for metacarpal thickness). The following variables were not significantly related to loss of spinal mineral: age, months after menopause, race (descent), height, weight, parity, smoking habit, calcium intake, and level of activity. The initial levels of all three skeletal measurements or the changes in the radius or metacarpal after 2 years did not correlate significantly with changes in spinal mineral content. In the 56 women not receiving estrogen, there was no relation between calcium intake and the spinal mineral loss after 2 years or between activity level and spinal mineral loss after 2 years (Figure 2). Estrogen use was the only independent variable showing a significant correlation in the stepwise multiple regression analysis. The 8 women who were given 0.625 mg of conjugated estrogen because of their low baseline spinal mineral values showed no significant loss of mean spinal or peripheral bone mineral content.

Discussion

Spinal mineral loss after spontaneous menopause was less than that we earlier observed after oophorectomy (1). Although some of the women who had spontaneous menopause in the present study had a high rate of spinal mineral loss, the mean annual rate of loss was 5% compared with nearly 9% in the oophorectomized women. Peripheral skeletal changes were also less marked—about half as great as we observed in the women with oophorec-

tomy.

Some residual ovarian function may have been present in our current series, but tests of follicle-stimulating hormone and estradiol substantiated menopausal status, and only one woman had a single episode of spontaneous bleeding after menopause. Her biopsy study showed endometrial atrophy. The differences could also be explained by the widely held belief that the rate of bone loss is greatest at the time of estrogen loss and the subsequent rate of loss diminishes with time. The women in our present study were 10 years older and already had had loss of ovarian function for a mean of 2 years. They might have had a higher rate of bone loss before entering our study, possibly even in the few years just before cessation of menses. Against this hypothesis is our inability to find any significant relation between the number of months after menopause and the annual rate of bone loss.

SELECTION BIAS

Could factors other than treatment account for the profound differences observed between groups in this study? If women discontinued treatment because of bone loss, the observed group means would have been biased toward a more beneficial effect. The three women receiving estrogen who had completed 1 year's study but elected not to have a 2-year study did not have failure of treatment: their spinal mineral changes at 1 year were -0.9% , -1.9% , and -2.4% , losses that were representative of their treatment group. Their reasons for dropping out were loss of interest and fear of further radiation. All have continued to use the same dose of estrogen. The three women dropping out of the calcium group were not different from those continuing: their spinal mineral changes seen after 1 year were -1.7% , -4.6% , and -14.5% , representative of their treatment group. Their reasons for dropping out were also lack of interest and fear of further bone mineral tests, and none was switched to other treatments.

Examination of other possible confounding variables

shows that women not receiving calcium had a greater (albeit not significant) number of some putative risk factors for osteoporosis: they were lighter, more inactive, and smoked more than calcium users. These factors could only increase the likelihood of finding a salutary effect of calcium. Women in the calcium-supplemented group were similar to the estrogen users but were a mean 3.6 kg heavier. Thus, no known risk factors were present that could bias results in favor of estrogen. With the exception of estrogen treatment, multivariate analysis was unable to show any factors that significantly predicted spinal mineral loss.

ESTROGEN DOSE RESPONSE

The dose-response relation for the osteotrophic effect of various estrogens has been evaluated previously, and we have shown that 0.6 mg/d of conjugated estrogens was required to protect oophorectomized women against bone loss (1). Our present study confirms the effectiveness of this dosage in women having had spontaneous menopause who have low initial spinal mineral content. Lindsay and associates (4) found 0.6 mg was effective, and Horsman and coworkers (3) showed that ethinyl estradiol, 0.02 mg, a dose equivalent to 0.6 mg of conjugated estrogens, was also the minimum protective dosage. Christensen and associates (2) found that 1 mg of estradiol-17 β , combined with 800 mg of calcium supplementation, was sufficient to maintain bone mineral.

Our study shows that 0.3 mg/d of conjugated estrogens is sufficient prophylaxis against bone loss, if it is combined with a daily intake of 1500 mg of calcium. We did not test 0.3 mg/d of estrogen without calcium supplements in this study, because we (in a study of oophorectomized women) (1) and others (in studies of both oophorectomized women and women having spontaneous menopause) (2-4) had previously shown that dosage to be ineffective.

PROGESTOGEN EFFECT

Could factors other than the added calcium account for the satisfactory preservation of bone mass seen with low-dosage estrogen treatment? An important consideration is concomitant progestogen therapy. Lindsay and associates (6) administered 19-nor-17 β -hydroxyprogesterone to ten menopausal women and found that metacarpal bone mineral content was preserved. The study was not continued beyond 1 year and may have lacked power to show significant losses due to its small number of subjects, brief period of follow-up, and use of a relatively insensitive bone measurement technique. Hart and coworkers (7) also studied two other progestogens, norgestrel and norethisterone, and found that only norethisterone, when given without estrogen, prevented loss of metacarpal bone mineral content over a 2-year period. No published data show the effects of progestogens on spinal mineral loss, and it is impossible to predict from peripheral measurements what the effect might be.

Each of these studies prescribed the progestogen daily, whereas we used it only 10 days a month. Three women in our study used 0.3 mg/d of estrogen without progesto-

gen. Their mean 2-year cumulative changes in bone mineral values showed +1.4% for spinal trabecular mineral content, +0.6% for radial mineral content, and -0.6% for cortical thickness. Although this group was small, it appears the progestogen was not necessary for the salutary effects of calcium with 0.3 mg of estrogen. These findings, in conjunction with the inconsistent and incomplete published data on progestogen alone, make doubtful any beneficial effect of the coadministered progestogen in our study.

CALCIUM EFFECT

The effect of calcium on bone mass has been examined in several epidemiologic and experimental studies. Bone mass was assessed in a cross-sectional epidemiologic study in Yugoslavia (8). Women living in a district of high dietary calcium intake were compared with women living in one of low-calcium intake (average intake, 1500 and 500 mg, respectively—the same range of most experimental studies). The bone mass before menopause, measured by metacarpal cortical area, was greater in women from the high-calcium district, but the loss seen with aging was also greater, resulting in a diminishing advantage with increasing age. Hip fractures were less common in older women in the high-calcium district, but wrist fracture prevalence did not differ. These findings suggest that high-calcium diets may retard the development of osteoporotic fractures by inducing a greater peak adult bone mass and not by retarding bone loss, and that calcium may not protect against bone loss in all parts of the skeleton.

Johnell and Nilsson (9) found no correlation between calcium intake and distal forearm bone mass in a cohort of 49-year-old women. In a cross-sectional study of women aged 20 to 80 years, Lavel-Jeantet and associates (10) found no relation between calcium intake and spinal trabecular mineral content; most of the women in the study had had menopause, but the results were not stratified by age. Kanders and associates (11), in a study of women aged 25 to 34 years, found that high-calcium diet combined with a vigorous exercise habit was associated with higher forearm bone mass, but that the effect of calcium alone was minimal. Anderson and Tylavsky (12) showed that women with higher lifetime consumption of calcium had more cortical bone in the midforearm but no effect of calcium intake was seen at the distal forearm, where there is some trabecular bone.

Heaney and coworkers (13) measured calcium balance in women before and after menopause and found that low calcium intake was associated with negative calcium balance, whereas high intake was associated with positive calcium balance. They suggested that 1500 mg/d would be sufficient to prevent bone loss in menopausal women. Recker and coworkers (14) showed that calcium supplements were partially effective in retarding menopausal bone loss when the treatment was started an average of 7 years after menopause. The metacarpal combined cortical thickness of women treated with calcium was preserved, whereas untreated women lost 2.4% of their initial values during the 2 years of study. Assessment of

forearm bone mineral content in these same women indicated that calcium treatment was ineffective: women taking calcium lost 3.7% of initial bone mass, a percentage not significantly different from the 5.8% loss seen in the untreated women. Horsman and coworkers (15) also reported that calcium supplements were effective in retarding bone loss when given to women an average of 7 years after menopause. Nilas and coworkers (16) studied 103 women in early menopause and concluded that high calcium intake had no protective effect against bone loss measured by photon absorptiometry at the distal radius. Mean rates of loss in women with low, medium, or high calcium intake were nearly identical. After 2 years, women with extremely low and extremely high intakes of calcium had similar rates of bone loss: 3.9% and 4.1%, respectively.

Our present study and the work of Nilas and colleagues (16) refute the claim that a high-calcium diet will prevent the rapid bone loss that occurs in early menopause. Calcium may be effective in preventing bone loss at skeletal sites other than the forearm or spine, and cortical bone may benefit preferentially. Additionally, calcium may retard or even prevent bone loss in older women. Nordin and associates (17) have suggested that calcium's benefits are greatest in women with calcium malabsorption, a condition that develops with increasing frequency after age 60 to 65 years.

ESTROGEN EFFECT

We recommend that women receiving estrogen replacement therapy be given the smallest effective dosage, because the incidence of side effects is dose related. Studies of endometrial abnormalities in women using estrogen have shown that hyperplasia, particularly the adenomatous and atypical variants, is more frequent at higher dosages of estrogen (18). To date, no studies of the endometrium-stimulating effects of the 0.3-mg/d dosage of estrogen have been reported, but it is reasonable to assume that endometrial stimulation and its attendant problems will be less than those caused by 0.625 or 1.25 mg/d. Christensen and associates (2) found a lower prevalence of withdrawal bleeding in women who used 1 mg of estradiol-17 β than in those who used 2 or 4 mg. We concur that at least half the women on low-dosage estrogen treatment do not have withdrawal bleeding.

Calcium supplements alone do not protect against the accelerated bone loss that occurs at menopause and that preferentially affects trabecular bone. We would expect that fractures of the wrist and vertebrae would not be prevented by administration of calcium alone to postmenopausal women. By the simple expedient of adding calcium to the diet, estrogen therapy may be made more safe and acceptable without loss of its protective effects on the skeleton. Before widespread adoption of this new treatment regimen, large-scale, long-term studies should confirm its efficacy.

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